

Students' Experience of a ChatGPT Enabled Final Exam in a Non-Majors Chemistry Course

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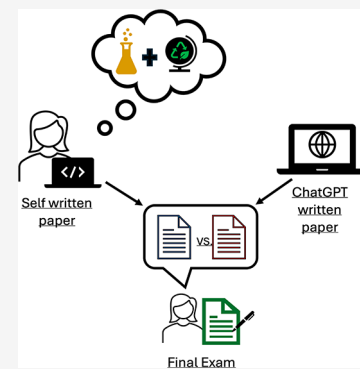
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ABSTRACT: In the field of education, ChatGPT has become a topic of debate for its usefulness as a learning tool. This article focuses on non-science majors' ($n = 29$) perceptions of a ChatGPT enabled final exam, where, prior to the exam, students wrote papers on science and sustainability and, during the final exam, students were asked to compare their paper to one produced on the same topic by ChatGPT. Thus, the underlying chemistry, its broader impacts, and connection to sustainability and writing styles were compared. Students' perceptions were analyzed through a developed coding framework that enabled the visualization of emerging themes. The most common themes revealed that students believed the ChatGPT essay did not read as "human-like", used more intricate words, and often did not include enough science to support its arguments. Students also noted that their essays provided more chemistry details and were easier to read as they focused on connecting chemistry concepts to their essay topic as well as sustainable policies and practices. Students were impressed, however, by ChatGPT's ability to discuss various sustainability solutions, policies, and practices. The final exam inspired self-reflection for the students to improve not only their writing but also their analysis of sustainability responses. Overall, students rated the comparative activity as a final exam to be favorable and remarked on the importance of analyzing AI generated work for the future of learning.

KEYWORDS: Student Writing, Science for Non-Majors, Generative AI, Writing Comparison



INTRODUCTION

The emergence of large language models (LLMs) for generative artificial intelligence (GAI) applications in writing has produced an array of reactions in academia.¹ One common reaction has been alarm based on the expectation that GAI writing tools such as ChatGPT could certainly pose challenges for maintaining academic integrity.² Chemistry and chemistry education is no exception to the observation of widespread reactions to this technology as is evidenced by the collection of a Virtual Special Issue related to GAI in teaching and learning chemistry.³

A number of articles investigating ChatGPT and its potential for chemistry education have appeared. Many of these efforts have focused on investigating the capabilities of such systems to write about chemical topics with accuracy. Investigations have considered ChatGPT performance on exam questions from courses such as general chemistry⁴ and pharmaceutical science.⁵ More specific content investigations with a focus on acid/base calculations⁶ and organic chemistry⁷ have also been reported. Notably critical evaluations of ChatGPT performance on chemistry tasks have also resulted in these broad efforts to probe content accuracy.^{8,9}

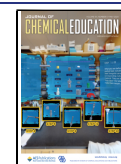
Although these attempts to articulate areas of potential concern provide a useful sense of caution with respect to the incorporation of GAI, ways that the technology can be incorporated into teaching and learning chemistry have also been reported. Recognizing the importance of writing in laboratory courses, the potential for GAI assistance in report writing has been studied¹⁰ and student perceptions of this emerging technological capacity have been probed.¹¹ This early snapshot of student views suggests that there may not be widespread use of the tool yet, and generally mixed views about its potential role in education.¹¹ A key study in terms of the project described here focused on mechanistic reasoning in organic chemistry has been carried out and presented a comparison of student and GAI written answers predicated on such reasoning within a writing-to-learn paradigm.¹² As has

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been noted in several other studies, there are specific aspects of the narrative needed to describe organic mechanistic reasoning where GAI does not yet appear to be sufficiently trained. Additional work with helping student writing and its connections to critical thinking using a several step assignment and regular feedback to students has also been studied and reported.¹³

In addition to these observations, a few articles have appeared that suggest avenues for enhancing student learning by leveraging GAI tools.^{14–16} In one case, several possible ways to incorporate GAI are suggested.¹⁴ Additional work has started to envision teaching and learning activities involving chemistry students, including practice in generating prompts for ChatGPT that advance a multistep writing and analysis project.¹⁵ Other authors have focused specifically on the potential to enhance metacognitive skills of students through the use of GAI tools.¹⁶ This study used the teaching of chemistry problem solving skills along with ChatGPT to engage students in an explicitly metacognitive exercise to improve those skills, along with critical thinking.

Considering this summary of reports on educational usage of GAI tools, there are effectively three types of activities. First, there are comparative studies to gauge the content accuracy of GAI. Second, multiple ideas of ways to begin work with GAI tools have been postulated, and comparison between GAI outputs and student outputs has been carried out by instructors. Third, student activities that leverage GAI have started to emerge as well, with an emphasis on enhancing desirable skills such as creative thinking and metacognitive awareness. The activity reported here best fits with this final category, where student learning activities have been engaged, and it also includes the concept of comparative efforts; only in this case the comparisons are made by the students themselves as explained below.

■ THE STRUCTURE OF PAIRED ACTIVITIES TO ENGAGE STUDENTS WITH GAI TOOLS

The course in which a GAI component has been added is imbued with themes of sustainability, with an overall perspective that such efforts are vitally important for helping students of chemistry contribute to the many challenges facing society. Considering the concept of planetary boundaries,^{17,18} for example, suggests a number of planetary systems that are at risk where chemistry will be needed to find sustainable solutions for the planet. Calls for incorporating sustainability into the teaching have been present for several years.^{18–23} One key challenge for such inclusion, however, has been the issue of assessment, and writing about chemistry and sustainability has emerged as a useful strategy.²⁴ Thus, for a course in chemistry for non-science majors taught in the fall 2023 semester, a writing assignment was assigned that had students connect the chemistry of carbon or water footprint to issues of sustainability. Such assignments have been commonplace in courses taught by the author team,²⁴ but in this case an additional GAI connection was included.

The Fall 2023 offering is the first time that this course has paired the writing assignment with the nature of the final exam. Students enrolled in this course ranged from their first year in college to seniors nearing graduation. Twenty-nine students were enrolled in the course, and IRB approval was acquired to use student artifacts from the course to analyze the effectiveness of this paired writing-final exam teaching intervention. Twenty-seven students gave consent for the use

of their papers for analysis. An additional question to judge student impressions of the final exam strategy was included with modest extra credit points for the final exam. All students were able to gain extra credit independent of whether they gave consent for use of their work. The writing assignment was due after roughly 2/3 of the course had been completed. This timing was designed so that (a) there was time to cover a range of related concepts in chemistry and (b) the paper would be graded for the students relatively close to the end of the semester and therefore be at least somewhat fresh in their minds. The main prompt for the paper was the following:

"For the purpose of the writing assignment we will focus on water or carbon containing molecules as the general topic. We will approach this idea by introducing some aspects of "systems thinking" and identifying how we can think about the ideas of "water footprint", "carbon footprint" in things we encounter in everyday life. The writing assignment itself may deal with either of these two concepts.

In your treatment of the topic you need to (1) identify which aspect (water footprint, carbon footprint) you are using in the paper; (2) Define the idea (water footprint, carbon footprint) and how it connects to chemistry in some way and (3) make a connection about how understanding this topic helps inform the idea of sustainability (you can do this by connecting to one of the UN Sustainably Development Goals, for example)."

The length of the paper was 750 words, not including references/bibliography. Course instructors often worked with students to help them refine their topic to be achievable in the relatively short word limit. A curated chatbot²⁵ was also available to students to answer questions related to the paper.

The idea that the student paper would be incorporated into the final exam along with an individualized ChatGPT version designed to be on the same topic was identified in the course syllabus and on the first day of the class. Thus, students in the course were aware that their final exam would be individually customized and consist of three things: (1) their paper as handed in and graded; (2) a ChatGPT version of the same paper; and (3) a set of questions asking them to compare the two versions of the paper on the topic of their choosing, within the broad outline of the chemistry and sustainability implications of either carbon footprint or water footprint.

Discussions about how the final exam would be created were held in-class for the last 4 meetings. Students knew going into the final exam that there would be 3 comparison questions asked. The idea that a comparison would be expected on topics related to chemistry, sustainability, and writing styles was discussed in broad terms. These discussions prior to the exam were desired because the format was a first for most or all of the students in the class, and suggestions for how they might plan their approach were designed to alleviate some of the anxiety associated with a new final exam experience. At the exam sitting, students could not have a computer open, but it was an open notes exam so they had access to their course notes if they wished.

In order to create the customized exam for each student, their writing assignment was first printed out so they would have a hard copy of that paper at the time of the final exam. The second piece of the exam was to have ChatGPT render a version of a paper on the same topic as that of the student. The ChatGPT version was constrained to be roughly the same length as the assignment. The course instructor generated the prompts, which typically included the aspect of water or carbon footprint, along with an indication that the paper

Table 1. Example Statements Showing Perceived Differences between the ChatGPT Paper and the Students' Paper with Regard to Writing

Code Category	Code Detail	Differences		Excerpt Examples
		Student Paper	ChatGPT Paper	
Structure	Writing	Essay-like ($n = 5$)	Bulleted list ($n = 8$)	"The way in which ChatGPT wrote the paper was more of a bulleted list type form whereas I wrote the paper in a more essay type form" - Student 1
	Words	Authentic ($n = 4$)	Artificial ($n = 4$)	"ChatGPT used more intricate words than me in some places, some of which don't sound 'person-like' and could give away that it's an AI generating work" - Student 5
	Organization and reading	Better organized ($n = 1$) and smoother reading ($n = 1$)	Easier to access ($n = 2$) and to read ($n = 1$)	"ChatGPT allows for easier access to information with little search but my paper allows for a smoother reading process as a whole. Both styles of writing work, just in slightly different ways." - Student 8
Information	Statistics	Included numbers ($n = 3$)	Lacking ($n = 4$)	"I was expecting the ChatGPT version to have more statistics and information than mine." - Student 3
	Depth	Thorough ($n = 2$)	Surface level ($n = 5$)	"Compared to ChatGPT, I wrote a more in-depth analysis of the science behind crop growth, and the effects of pollutants on freshwater and the environment." - Student 12
Reasoning	Supporting claims	Provided more evidence ($n = 3$)	Focused on solutions ($n = 3$)	"ChatGPT failed to give any scientific context behind...and didn't provide any numerical data to support its reasoning..." - Student 10

should be 750 words and include discussions of chemistry and of sustainability. For example, several students chose to consider the water footprint of the fashion industry (or fast fashion) in some way and a prompt used to produce the ChatGPT comparison paper for such a choice could be "Write a 750 word essay on the water footprint of clothing production and its impact on sustainability." In roughly five cases ChatGPT went over the word limit, and editing for length was carried out before the ChatGPT versions were printed out. Finally, the test itself was created with three questions/prompts:

1. Compare your writing with the writing of ChatGPT.
2. Compare the information about science you included in your essay with the information about science that ChatGPT put together.
3. Compare the information about sustainability you included in your essay with the information about sustainability that ChatGPT put together.
4. (Optional extra credit) How would you rate the concept of comparing your essay with one by ChatGPT as a way to help learn about the topic (water footprint or carbon footprint) you did?

The set of three documents were paper clipped together, and each student was handed their own custom final exam at the start of the 2 h assigned time period. Most students finished in around 90 min. The prior discussions about possible strategies for the in-test writing comparing the two essays appeared to have provided useful guidance, as the exam answer essays were all germane to the assigned comparative tasks. Grading of the exams was completed by the course instructor, and it is worth noting that the general impression of a high quality work was evident. The IRB protocol for this evaluation of a teaching innovation does not allow comment here on score ranges for the student performances on the test.

■ PREPARATIONS AND METHODS OF ANALYSIS OF STUDENT WORK AND IMPRESSIONS

Given the nature of the final exam, the artifacts that were collected were the student, handwritten papers with answers to the 3 questions and the extra credit question related to their impressions of the use of ChatGPT as part of the final. Student responses were anonymized prior to any analysis being carried out, and the handwritten exam answers were typed. No student names or potentially identifying demographic information was

kept on the typed student answer documents. All information was listed as arising from "Student #".

All student answers to optional question 4 were captured. For each of questions 1–3, it was apparent that trends in student impressions and responses were identifiable with smaller numbers of students. With 27 students allowing the use of their papers, three independent groups of 9 students were randomly assigned to capture answers for the three different paper comparison questions.

A two-phase coding approach was utilized to code students' answers. First, an educationist approach was used, followed by a more holistic approach to identify emerging themes. Cresswell and Poth's data analysis spiral²⁶ informed the coding process. This approach involves managing and organizing the data, familiarizing the data accompanied by memo making, and describing and classifying the codes through two coding cycles in this case for each question analyzed. Interpretations are then developed from the codes, and finally, representations and visualizations are produced. The two coding cycles continued until satisfactory codes were developed by two coders, who engaged in a negotiated agreement after coding each question and after completing each cycle. This approach involved back-and-forth communication between the coders regarding the meaningful excerpts that were selected for coding and the codes used to describe the excerpts. The codebook was developed inductively during the first coding cycle and was used for deductive coding during the second cycle. According to Saldaña,²⁷ this approach solidifies the coding system before it is used deductively for further data analysis. The combined coding approach and negotiated agreement aided in developing themes to better understand students' perceptions as they compared their writing to that of ChatGPT.

■ RESULTS AND DISCUSSION

Structure, Information, and Reasoning

Students perceived that the writing in the ChatGPT paper differed most notably from theirs on the basis of structure, information, and reasoning. Students referred to differences in the writing structure, word choice, and ease of reading that were often associated with the organization of content. They also claimed that the depth of information and use of statistics differed, which they contrasted from a reasoning perspective. The categories, codes, and key differences evident in students' answers are outlined in Table 1.

Repeated perceptions from students' comparisons indicated that their papers were more essay-like, which contributed to a better flow with greater coherence and connections between their ideas as opposed to the "bulleted list" format that was produced by ChatGPT. Responses of students 1 and 8 in Table 1 exemplify these sentiments. Additional support of this observation from Student 2 stated: "while mine was in paragraph form, ... [ChatGPT] didn't use connecting setup sentences to connect each topic to make it flow." It must be noted at this point that these students likely have more exposure to writing assignments with an essay-like structure and thus have developed skills based on their prior experiences. It is also possible that some of the students have never written a science essay before and, thus, had no expectations about the structure of the paper. Indeed, one student expressed this very thought process when they commented: "I'm wondering if this is how science essays are supposed to look, seeing as I've only written essays for English classes before." - Student 5.

Students expressed different opinions with regard to the writing structure they preferred. As shown in Table 1, Student 8 noted that the ChatGPT paper allowed for easier access to information and reading, whereas others felt their papers were easier to read due to the smooth transitions, informal structure, and word choice. Students experienced the formal writing style and professional word choice of ChatGPT to be "robot-like" and stated that the "more intricate words" used by ChatGPT do not sound "person-like". For example, student 5 thought that "ChatGPT used more intricate words than me in some places, some of which don't sound "person-like" and could give away that it's an AI generating work." There was not a shortage of opinion on preferences; however, the overall consensus was stated well by Student 8 who experienced "both styles of writing work, just in slightly different ways."

Several students also perceived that they included more statistics to support their claims and that ChatGPT lacked numerical evidence in support of its arguments. Student 12 stated that "ChatGPT is devoid of statements that provide evidence to make a compelling scientific claim." In agreement with this claim, Student 3 also stated "I was expecting the ChatGPT version to have more statistics and information than mine." It was evident from students' answers that they gave credit to ChatGPT for its ability to cover a broad range and variety of topics. Student 26 mentioned, "I think ChatGPT has the advantage of pulling all possible data, in this case, detailed problems and solutions, straight from its database. It was able to touch on multiple things I did not." As a large language model, ChatGPT can obtain information from various sources, but students still perceived that their writing was more focused, especially on the science and chemistry content.

Comparison of Science and Chemistry Information: What Does "Science" Mean?

According to the second question on their final exam, students were asked to compare the information presented about science in their own writing to the information about science in ChatGPT's writing. Many students perceived that ChatGPT included more information about science, specifically, more chemistry details, to support their claims. Table 2 includes the main differences that students found between the science in their paper and the science in ChatGPT's paper. The key differences were related to the connections among science, context, impact, and sustainability as well as chemistry details and the important role chemistry plays in their everyday lives.

Table 2. Example Statements Showing Perceived Differences between the ChatGPT Paper and the Students' Paper with Regard to Science

Code	Connection to	Differences			Excerpt Examples
		Student Paper	ChatGPT Paper		
	Context	Explains science (n = 3)	Lacking depth of science (n = 4)		"...my paper gives more information about the science behind carbon emissions and a carbon footprint than ChatGPT's paper... ChatGPT's paper failed to go further in depth with the science behind Amazon's carbon footprint" - Student 16
	Impacts	Includes microscopic cause of the impact (n = 5)	Lacks depth of science (n = 4)		"ChatGPT also doesn't mention the issues of particulate matter along with the health risks. ChatGPT lacks including the respiratory problems humans face with particulate matter." - Student 16
	Sustainability	Calls for sustainable action (n = 1)	Provides a variety of sources (n = 1)		"I feel as if the entire sustainability section belongs with the science. I spent the last paragraph or two showing the reader ways in which they can do their part in reducing carbon... However, I did appreciate the information ChatGPT included such as cotton being very water intensive, very poor working conditions in factories, and waste having an impact on our ecosystems." - Student 22
Chemistry	Detail	More (n = 4)	Less (n = 2)		"tried to explain why water is so important in manufacturing using chemistry, discussing its high heat capacity and classification as a solvent." - Student 10
	Importance	Role and relevance discussed (n = 2)	Not discussed (n = 1)		I touched base on how food is central to carbon-containing molecules, this demonstrates the chemistry that shapes our daily lives. - Student 11

Students perceived ChatGPT's writing as more solution focused rather than science based and argued that the information presented should have science to back up the claims, as demonstrated by student 16 in Table 2. Student 15 even went so far as to say "I am proud to say that my paper gives more information about the science behind carbon emissions and a carbon footprint than ChatGPT's paper." However, Student 18 mentioned that "[ChatGPT's paper and the student's paper] both are similar and cover the same things. They just have different wording and ChatGPT sounds more artificial in a way but also covered more specific topics like feed production and deforestation." These differing statements about how much science was included can be attributed to student differences as well as how each student perceives "science or chemistry" content.

The question then becomes, how do students and ChatGPT differentiate what can be considered as science content? Several students perceived their writing to be more detail oriented, like Student 10 in Table 2 and Student 13, stating "My paper discussed how greenhouse gas molecules are shaped in a way that allows them to interact with IR radiation. IR radiation reacts with CO₂ and emits again in all directions. ChatGPT failed to provide reasons why carbon dioxide traps excess heat, heating the planet, furthering climate change." These students found it important to mention the chemical processes that are behind specific phenomena and emphasize the role that chemistry plays in everyday life. Regarding the efforts from ChatGPT, students believe that its paper loosely relates chemistry to context, lacking in depth to contribute compelling statements.

Additionally, students tended to express disappointment in the science portion of the ChatGPT paper. Student 15 stated that they "do wish ChatGPT's paper would have explained more of the science behind carbon emissions. I think it would have been interesting to see if its scientific explanations were easy to comprehend as my own were. I also wonder how accurate ChatGPT would be with scientific reasoning or if the sources it uses are even reliable." This statement includes the idea that most students experienced the taking of this final as a learning experience. When there was a lack of new information gained from ChatGPT's paper, it seemed to hurt the credibility of ChatGPT as a useful tool for students.

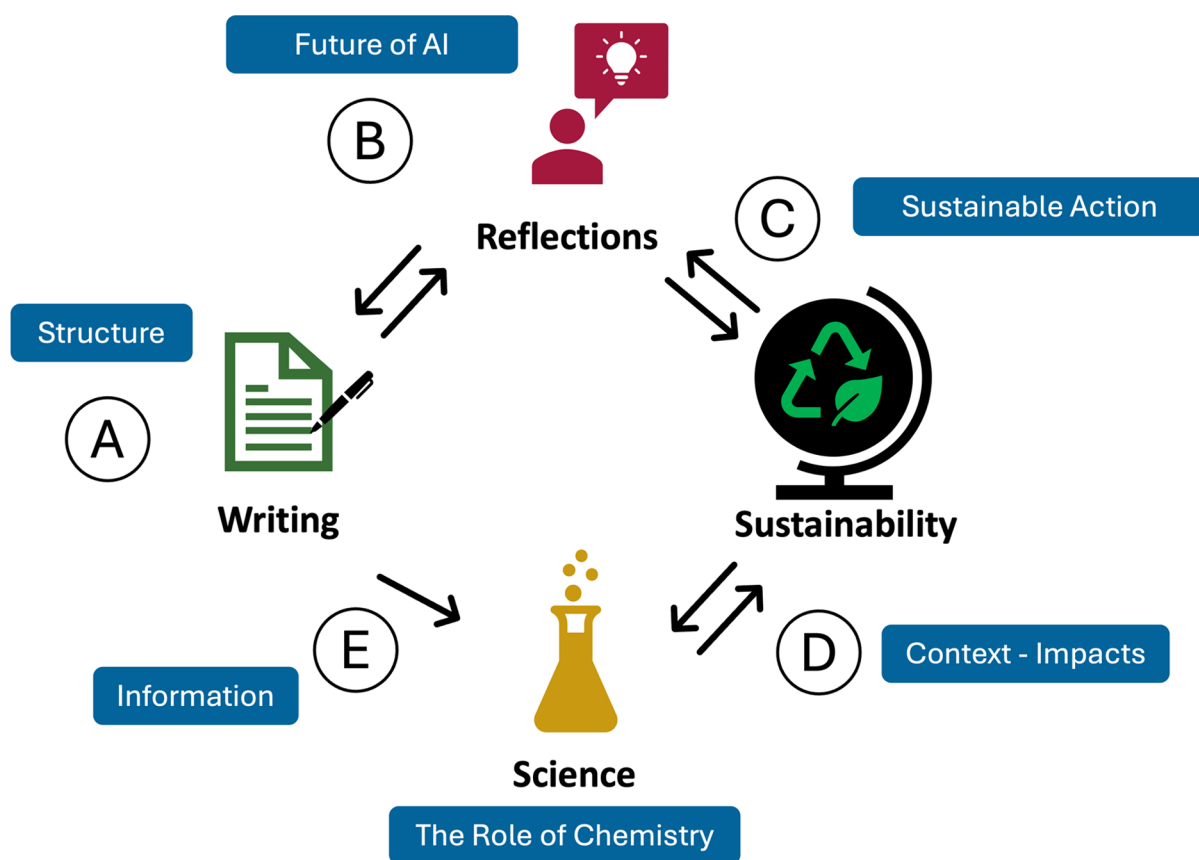
Perceptions of Comparing Sustainability

In question 3, students were prompted to make connections between how their understanding of the carbon and water footprint can help inform sustainability as it relates to the UN Sustainable Development Goals. Students' answers, as summarized in Table 3, revealed various connections to chemistry, its role in the carbon and water footprint, potential implications in society, the economy, and the environment, and ultimately the influence of these dynamic relationships on sustainability. It was evident that in both papers most topics were centered around sustainable action and included goals to achieve future sustainability, the implementation of policies, support, and practices, and the challenges that come with implementing these changes.

Students perceived their sustainability content compared well to that of ChatGPT's paper. One student described that "both papers started off by identifying sources that contribute to ones carbon footprint in Oceania. Both papers talked about urbanization, industrialization, and uses for energy production. I also noticed that both papers concluded with the need for more urgent changes." Students also noted that both papers referred to sustainable action in the sense of goal setting and the UN

Table 3. Example Statements Showing Perceived Differences between the ChatGPT Paper and the Students' Paper with Regard to Sustainability

Category	Code	Differences		Excerpt Examples
		Student Paper	ChatGPT Paper	
Connections to Sustainable action	Impacts and context	Less variety (n = 2)	More variety (n = 2)	"It [ChatGPT] was able to touch on multiple things I did not, such as the sustainability implications of retail's transportation and packaging of goods." - Student 26
	Challenges	Lacking (n = 1)	More (n = 2)	"The AI bot also gave challenges golf courses may face when trying to implement these solutions ... I thought it was a pretty neat thing to add. Overall I think ChatGPT did a great job giving info about sustainability and problems which may come with that." - Student 25
	Goals	Personal and local (n = 2)	Societal and global (n = 3)	"Both essays talk about different things that affect our goal for sustainability and how we can apply our knowledge to pave the way for sustainability." - Student 19
	Policies	Few given (n = 1)	More given (n = 3)	"It [ChatGPT] also mentions different factors that can help us reach sustainability which are socially inclusive fair policies, public support, and practices that contribute to society's well-being in favor of sustainability." - Student 19
	Support and practices	Few examples (n = 5)	Exhaustive list (n = 6)	"I honestly believe ChatGPT gave better ways of becoming sustainable. I talked more about how to be sustainable with already made clothes ... ChatGPT went over that way of sustainability but also other ways." - Student 20



- A** The [ChatGPT] paper gave brief examples of **why water is important** in the fashion industry, such as its role in cotton production, **but provided no numbers to support its claims.** – student 10
- B** “ChatGPT used **more intricate words** than me in some places, some of which **don’t sound “person-like”** and could give away that it’s an **AI generating work**”. Student 5
- C** “I think it is really **impactful** in our generation to learn and work with/about ChatGPT and **recognize its strengths and weaknesses.**” student 26
- D** “I feel as if the entire **sustainability section belongs with the science.** All are important to mention and further give the reader **background** on why the reader **should care about the issue.**” student 22
- E** “Now going back to my essay. We jump right back into the **scientific side of achieving sustainability**... these principles guide us toward **more sustainable practices.**” student 11

Figure 1. Graphic demonstrating the main themes (structure, information, context, the role of chemistry, the future of AI, and sustainable action), with regard to each final exam question and how they are all connected.

Sustainable Development Goals (UNSDGs). One of the examples to illustrate this was expressed by Student 21, who stated that “*sustainability is where I believe our versions are the most similar. Not only did we both pull ideas from the United Nations sustainability goals, but we also had some of the same examples.*”

The difference in both papers as noted by many students was that ChatGPT provided a wider variety of ways to work toward sustainability, which include implementing support, practices, and policies. ChatGPT’s ability to pull together all of these solutions also left a good impression on students. As one of the examples, Student 25 commented “*The ChatGPT essay definitely blew me out of the water when it came to writing info*

about sustainability. While I only gave a few ideas to increase sustainability...the chatbot gave me a long list of other solutions." Student 20 also stated "I honestly believe ChatGPT gave better ways of becoming sustainable.... ChatGPT went over that way of sustainability but also other ways." Student 25 also noted that in comparison to their paper, "the AI bot [CHATGPT] also gave challenges ... when trying to implement these solutions."

There was also a mention that some of the suggested solutions provided by ChatGPT were unrealistic or irrelevant to their context. Student 23 expressed that "both of the papers mentioned zero-emission or electric cars, but as I mentioned above, it is not realistic for me and may not be for other college students." However, for Student 23 the ChatGPT paper inspired them to consider the impacts of their actions on a global scale rather than just locally, stating "the ChatGPT essay was a great choice as lowering ISU's footprint can make a major impact on Ames's footprint as a whole." It was also evident that some students were encouraged to reflect on their role in sustainable action. Student 19, for example, acknowledges the importance of their role, mentioning that "both essays talk about different things that affect our goal for sustainability and how we can apply our knowledge to pave the way for sustainability." Despite the variance in perceptions concerning the connections made to sustainability, one student summarized a key broad observation about this exercise for a final exam, which is that "both papers are good in their own way, and I think we both mentioned the same amount about sustainability just in different ways." - Student 27

Student Views on a ChatGPT Enabled Final Exam

The four questions that students were asked in their final exam about comparing writing, science, and sustainability as well as reflecting on the exam as a whole are represented as icons in Figure 1. The writing structure and information, the role of chemistry, and its connections to context with broader implications together with sustainable action were some of the key ideas that emerged from the analysis of questions 1–3. Furthermore, during the reflective process, students also expressed their views on how they rated the exam and made comments about the future of AI. On average, the students rated the activity as 8.6 out of 10. Many students spoke positively about comparing their essay to ChatGPT's essay. Student 1 stated "I really loved the concept. I appreciate the inclusion of ChatGPT." Student 23 stated "I thought it was a creative way to do an exam and I believe it concluded my semester effortlessly." Additionally, the comparative activity served as a metacognitive exercise as Student 24 mentioned that they were able to "...learn about new topics, in regard to carbon footprint" and Student 1 learned "about sustainable options for prisons by evaluating the ChatGPT paper." This exercise allowed students to think about how they could make their papers better with this new information as well as analyze the components they completed well. Students maintain that they benefited from the comparative activity, especially since "it's also good not to have to study for some big scary exam!" - Student 26.

More praise came for the comparative activity as a final exam because of how it related to the future of AI. Student 26 commented "I think it is really impactful in our generation to learn and work with/about ChatGPT and recognize its strengths and weaknesses" and student 9 mentioned "I feel like it was a very unique and interesting way to see what I could have included in my paper. It also is a good use of new technologies that are

emerging which is important for me as I study to become a teacher."

Students recognize that AI has a place in the world and will likely have a place in their future classrooms, so it is important to learn about AI generated materials now rather than later.

Though many students commented positively on the comparative assignment, a few students expressed concerns about this activity as a final exam for various reasons. Student 15 said "honestly [I] didn't learn more about a carbon footprint through ChatGPT because it failed to explain scientific information", while student 21 said "I don't feel like I would rate this method very high. I don't feel like this feels much like a final and more like a test or project that would be mid semester. I also feel like this method doesn't work at least in this class because there our other tests weren't at all like this." A few other students make the point that there may need to be changes to the exam, such as a better definition of science, to be more effective as a learning tool.

Overall, positive reviews for the ChatGPT enabled final exam were expressed by a large majority of the students. We argue that these key ideas that emerged from students' comparative analysis, shown in Figure 1, provide a snapshot of the interconnection between the science behind water and carbon footprint and the broader implications on sustainability. Students also expressed that science is a fundamental aspect of sustainability that can inform future science practices and sustainable actions. Figure 1 echoes this perception about science and sustainability, demonstrating a cyclic process that involves continuous reflection to understand humans' roles in sustainable action policies, practices, and solutions. Although not explicitly noted by students, the fact that their writing project was reconnected with the course rather than just being one score in the gradebook appears to have been a positive experience.

SUMMARY AND FUTURE IMPLICATIONS

This article analyzes student responses to the four questions on a 2 h final exam where students received customized exams that allowed them to compare their writing, science content, and sustainability content to that of ChatGPT's. When comparing their writing, students remarked on the difference in the writing structure and the use of supporting claims in both papers. Students varied in their claims of which writing style was better than the other but notably agreed on the importance of using claims, statistics, and scientific facts to back up arguments made in the paper. The students' perception of how they defined science was challenged as they perceived the chemical processes and details to be integral in explaining why, for example, greenhouse gases are harmful or the effects of ocean acidification on marine life.

Students commended ChatGPT for its mention of a variety of sustainable practices. They emphasized how both papers (the students' and ChatGPTs') urge the reader to consider more sustainable practices, but students frame this as a personal challenge while ChatGPT writes these practices as a societal and global call to action. Overall, a majority of students enjoyed the activity, appreciating it as an alternative to taking a traditional final exam. Additionally, students expressed utilizing AI and learning more about what it can do as important steps moving forward. They recognize that technology is here for the foreseeable future, and we must learn as much as we can about it.

This comparative activity was implemented in a class of 29 students for a final exam. This activity occurred in two stages: (1) a midsemester write-to-learn essay and (2) the final exam that required a comparison of ChatGPT's paper to the student's previously completed work. This activity fosters learning in a way that connects core chemistry concepts to timely world events while providing a metacognitive experience for students to practice and learn from their scientific writing. The extent of grading time tends to place a constraint on this activity, particularly because final exams normally occur a short time before final grades are turned in. Investigating ways to incorporate this strategy for using GAI in larger courses remains a goal for upcoming work.

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Notes

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■ NOTE ADDED AFTER ASAP PUBLICATION

There was an error in Figure 1 in the version that was published ASAP on April 12, 2024. This was corrected in the version published ASAP on April 15, 2024.

The advertisement features a vertical strip on the left showing a 3D molecular model with grey spheres and colorful translucent surfaces. The main background is dark blue. Text is in white and yellow. The CAS logo is at the bottom right.

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